

Minimum Viable Hypothesis: Meta-Learning Adaptive Implicit Communication Protocols for Emergent Multi-Agent Collaboration

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Compute: 1 GPU

Hours: 5

Memory: 24 GB

1 Claim

A Meta-AICP agent, employing a lightweight module meta-trained to infer and adapt its interpretation of a partner’s communication protocol (P-map) from $K=3$ distinct, pre-defined types within the first few interactions, will achieve an average zero-shot 2-player Hanabi game score that is at least 1.0 point higher and statistically significantly ($p \leq 0.05$) greater than a FixedGeneralistICP agent. The FixedGeneralistICP agent uses a single, fixed interpretation strategy optimized for average performance across the same K partner P-map types. This tests if meta-learned P-map adaptation offers benefits over static generalization even in a highly simplified setting.

2 Dataset

- **Name:**

2-Player Hanabi Environment (Hanabi Learning Environment or similar).

- **Modification:**

The environment will be augmented with two explicit ‘scouting actions’ (SA1, SA2) available to each player, costing minimal or no game resources.

Partner agents will be scripted to use one of $K=3$ distinct, pre-defined, simple rule-based P-maps at the start of each episode. These P-maps define distinct meanings for sequences of SA1/SA2 to convey one of three specific, game-relevant pieces of information about the partner’s hand.

For example:

- **P-Map Type Alpha:** 'oldest card playability';
- **P-Map Type Beta:** 'oldest card is high-value (e.g., a 5)';
- **P-Map Type Gamma:** 'oldest card is safely discardable'.

The Meta-AICP agent will be meta-trained with knowledge of these K=3 P-map types as possibilities.

- **Evaluation:**

Zero-shot performance: measured on the first full episode of interaction with a partner whose P-map type is drawn from the K=3 set but unknown to the agent at episode start. Averaged over at least 1000 evaluation episodes per agent type, with diverse shuffles and partner P-map selections.

3 Metric

Average zero-shot Hanabi game score (0-25 points for 2-player). Statistical significance of the difference will be assessed using a t-test.

4 Baseline

FixedGeneralistICP: A reinforcement learning agent (e.g., using a standard Hanabi-compatible algorithm like R2D2, VDN, or a simpler off-policy/on-policy actor-critic) trained with a single, fixed policy and P-map interpretation strategy.

This agent is trained against the same population of K=3 P-map-using scripted partners, optimizing its fixed strategy for average performance across these partner types. It does not adapt its interpretation strategy within an episode.

5 Success Threshold

The Meta-AICP agent achieves an average zero-shot game score at least 1.0 point higher than the FixedGeneralistICP baseline, AND this difference is statistically significant ($p \leq 0.05$). A 1.0 point improvement on the 0-25 scale is considered practically meaningful in 2-player Hanabi. budget.